DATA ENGINEERING INTERVIEW PREP

100 DSA Questions for Mid-Senior Data Engineer Interviews

Easy-Medium Only | Curated for Data Engineer-relevant DSA topics

After interviewing for mid-senior Data Engineer roles at tier-1 product companies, one thing is clear: after SQL, Easy-to-Medium DSA still plays an important role in the first two technical rounds.

100 Curated Questions - Easy/Medium problems across 10 core topic areas.
Each question includes a direct practice link for focused interview preparation.

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What Was Included - and What Was Filtered Out

This practice sheet focuses on DSA categories that have repeatedly appeared in mid-senior Data Engineer technical rounds.

Included topics	Array, String, Stack, Queue, Searching, Sorting, Hashing, Matrix, Binary Search Tree, Basic Dynamic Programming
Question selection	Only Easy/Medium questions within the included topic categories are selected for practice.
Not included	Hard questions and unrelated categories such as Linked List, Graph, Heap, Trie and Bit Manipulation.
Question links	Each row contains a clickable direct link to its LeetCode problem page.

How to Practise

01	First pass Solve category-wise and understand the core pattern.
02	Interview pass Explain brute force, optimum approach, complexity and edge cases before coding.
03	Revision pass Re-solve incorrect or slow attempts after 3-5 days.
04	Mock pass Practise 2 SQL problems + 1 DSA problem under a timer.

Topic Coverage at a Glance

Exactly 100 Easy/Medium problems, grouped around the topic areas relevant to the intended interview preparation.

Topic	Questions	Starter Set	ocus
01. Array	10	7	Two pointers, subarrays, prefix products and in-place operations.
02. String	10	7	Character frequency, palindrome logic and sliding window patterns.
03. Stack	10	1	Balanced delimiters, nested parsing and monotonic stack reasoning.
04. Queue	10	0	FIFO design, scheduling simulation and BFS-style processing.
05. Searching	10	2	Binary search boundaries, rotated arrays and answer-space search.
06. Sorting	10	3	Ordering, partitioning and interval consolidation.
07. Hashing	10	3	Sets, frequency maps, constraints and prefix remainder logic.
08. Matrix	10	4	2D traversal, transformations, markers and grid backtracking.
09. Binary Search Tree	10	3	BST ordering, pruning, iterative traversal and updates.
10. Basic Dynamic Programming	10	9	State definition, recurrence, memoisation and tabulation.

01. Array

Interview focus: Two pointers, subarrays, prefix products and in-place operations.

10 curated Easy-Medium practice questions

01	Two Sum Pattern: Hash map lookup	EASY **PRACTICE**	**SOLVE >**
02	Best Time to Buy and Sell Stock Pattern: Running minimum	EASY **PRACTICE**	**SOLVE >**
03	Maximum Subarray Pattern: Kadane's algorithm	MEDIUM **PRACTICE**	**SOLVE >**
04	Product of Array Except Self Pattern: Prefix and suffix	MEDIUM **PRACTICE**	**SOLVE >**
05	Maximum Product Subarray Pattern: Track min and max	MEDIUM **PRACTICE**	**SOLVE >**
06	3Sum Pattern: Sort + two pointers	MEDIUM **PRACTICE**	**SOLVE >**
07	Container With Most Water Pattern: Two pointers	MEDIUM **PRACTICE**	**SOLVE >**
08	Move Zeroes Pattern: In-place traversal	EASY **PRACTICE**	**SOLVE >**
09	Remove Duplicates from Sorted Array Pattern: Slow and fast pointers	EASY **PRACTICE**	**SOLVE >**
10	Subarray Sum Equals K Pattern: Prefix sum + map	MEDIUM **PRACTICE**	**SOLVE >**

Revision action: Select any 3 questions from this topic and solve them again without viewing your earlier solution.

02. String

Interview focus: Character frequency, palindrome logic and sliding window patterns.
10 curated Easy-Medium practice questions

01	Valid Anagram Pattern: Frequency count	EASY **PRACTICE**	**SOLVE >**
02	Valid Palindrome Pattern: Two pointers	EASY **PRACTICE**	**SOLVE >**
03	Longest Substring Without Repeating Characters Pattern: Sliding window	MEDIUM **PRACTICE**	**SOLVE >**
04	Longest Repeating Character Replacement Pattern: Window + frequency	MEDIUM **PRACTICE**	**SOLVE >**
05	Group Anagrams Pattern: Canonical key	MEDIUM **PRACTICE**	**SOLVE >**
06	Longest Palindromic Substring Pattern: Expand around centre	MEDIUM **PRACTICE**	**SOLVE >**
07	Palindromic Substrings Pattern: Centre expansion	MEDIUM **PRACTICE**	**SOLVE >**
08	Reverse Words in a String Pattern: Token reversal	MEDIUM **PRACTICE**	**SOLVE >**
09	String Compression Pattern: Two pointers	MEDIUM **PRACTICE**	**SOLVE >**
10	Permutation in String Pattern: Fixed-size window	MEDIUM **PRACTICE**	**SOLVE >**

Revision action: Select any 3 questions from this topic and solve them again without viewing your earlier solution.

03. Stack

Interview focus: Balanced delimiters, nested parsing and monotonic stack reasoning.
10 curated Easy-Medium practice questions

01	Valid Parentheses Pattern: Matching stack	EASY **PRACTICE**	**SOLVE >**
02	Min Stack Pattern: Auxiliary minimum state	MEDIUM **PRACTICE**	**SOLVE >**
03	Backspace String Compare Pattern: Stack simulation	EASY **PRACTICE**	**SOLVE >**
04	Baseball Game Pattern: Operation stack	EASY **PRACTICE**	**SOLVE >**
05	Remove All Adjacent Duplicates In String Pattern: Collapse stack	EASY **PRACTICE**	**SOLVE >**
06	Evaluate Reverse Polish Notation Pattern: Expression stack	MEDIUM **PRACTICE**	**SOLVE >**
07	Daily Temperatures Pattern: Monotonic stack	MEDIUM **PRACTICE**	**SOLVE >**
08	Asteroid Collision Pattern: Collision simulation	MEDIUM **PRACTICE**	**SOLVE >**
09	Decode String Pattern: Nested stack	MEDIUM **PRACTICE**	**SOLVE >**
10	Simplify Path Pattern: Path stack	MEDIUM **PRACTICE**	**SOLVE >**

Revision action: Select any 3 questions from this topic and solve them again without viewing your earlier solution.

04. Queue

Interview focus: FIFO design, scheduling simulation and BFS-style processing.
10 curated Easy-Medium practice questions

01	Number of Recent Calls Pattern: Rolling queue	EASY PRACTICE	SOLVE >
02	Implement Queue using Stacks Pattern: Amortized queue	EASY PRACTICE	SOLVE >
03	Implement Stack using Queues Pattern: Queue rotation	EASY PRACTICE	SOLVE >
04	Time Needed to Buy Tickets Pattern: Queue simulation	EASY PRACTICE	SOLVE >
05	Number of Students Unable to Eat Lunch Pattern: Queue counting	EASY PRACTICE	SOLVE >
06	Design Circular Queue Pattern: Circular buffer	MEDIUM PRACTICE	SOLVE >
07	Dota2 Senate Pattern: Two queues	MEDIUM PRACTICE	SOLVE >
08	Reveal Cards In Increasing Order Pattern: Index queue	MEDIUM PRACTICE	SOLVE >
09	Design Front Middle Back Queue Pattern: Deque balancing	MEDIUM PRACTICE	SOLVE >
10	Rotting Oranges Pattern: Multi-source queue BFS	MEDIUM PRACTICE	SOLVE >

Revision action: Select any 3 questions from this topic and solve them again without viewing your earlier solution.

05. Searching

Interview focus: Binary search boundaries, rotated arrays and answer-space search.
10 curated Easy-Medium practice questions

01	Find Minimum in Rotated Sorted Array Pattern: Pivot binary search	MEDIUM **PRACTICE**	**SOLVE >**
02	Search in Rotated Sorted Array Pattern: Modified binary search	MEDIUM **PRACTICE**	**SOLVE >**
03	Binary Search Pattern: Classic binary search	EASY **PRACTICE**	**SOLVE >**
04	Search Insert Position Pattern: Lower bound	EASY **PRACTICE**	**SOLVE >**
05	First Bad Version Pattern: First true boundary	EASY **PRACTICE**	**SOLVE >**
06	Find First and Last Position of Element in Sorted Array Pattern: Dual boundary search	MEDIUM **PRACTICE**	**SOLVE >**
07	Find Peak Element Pattern: Slope binary search	MEDIUM **PRACTICE**	**SOLVE >**
08	Search a 2D Matrix Pattern: Flattened search	MEDIUM **PRACTICE**	**SOLVE >**
09	Koko Eating Bananas Pattern: Answer-space search	MEDIUM **PRACTICE**	**SOLVE >**
10	Capacity To Ship Packages Within D Days Pattern: Answer-space search	MEDIUM **PRACTICE**	**SOLVE >**

Revision action: Select any 3 questions from this topic and solve them again without viewing your earlier solution.

06. Sorting

Interview focus: Ordering, partitioning and interval consolidation.
10 curated Easy-Medium practice questions

01	Merge Intervals Pattern: Sort + merge	MEDIUM **PRACTICE**	**SOLVE >**
02	Non-overlapping Intervals Pattern: Greedy by end time	MEDIUM **PRACTICE**	**SOLVE >**
03	Insert Interval Pattern: Ordered interval merge	MEDIUM **PRACTICE**	**SOLVE >**
04	Merge Sorted Array Pattern: Reverse two pointers	EASY **PRACTICE**	**SOLVE >**
05	Squares of a Sorted Array Pattern: Two-pointer ordering	EASY **PRACTICE**	**SOLVE >**
06	Relative Sort Array Pattern: Counting or custom sort	EASY **PRACTICE**	**SOLVE >**
07	Sort Colors Pattern: Three-way partition	MEDIUM **PRACTICE**	**SOLVE >**
08	Sort an Array Pattern: Merge sort	MEDIUM **PRACTICE**	**SOLVE >**
09	Largest Number Pattern: Custom comparator	MEDIUM **PRACTICE**	**SOLVE >**
10	H-Index Pattern: Sort + threshold	MEDIUM **PRACTICE**	**SOLVE >**

Revision action: Select any 3 questions from this topic and solve them again without viewing your earlier solution.

07. Hashing

Interview focus: Sets, frequency maps, constraints and prefix remainder logic.
10 curated Easy-Medium practice questions

01	Contains Duplicate Pattern: Hash set	EASY **PRACTICE**	**SOLVE >**
02	Longest Consecutive Sequence Pattern: Set sequence scan	MEDIUM **PRACTICE**	**SOLVE >**
03	Top K Frequent Elements Pattern: Map + bucket/heap	MEDIUM **PRACTICE**	**SOLVE >**
04	Ransom Note Pattern: Frequency map	EASY **PRACTICE**	**SOLVE >**
05	Happy Number Pattern: Seen-state set	EASY **PRACTICE**	**SOLVE >**
06	Intersection of Two Arrays Pattern: Set intersection	EASY **PRACTICE**	**SOLVE >**
07	Unique Number of Occurrences Pattern: Frequency uniqueness	EASY **PRACTICE**	**SOLVE >**
08	Valid Sudoku Pattern: Constraint sets	MEDIUM **PRACTICE**	**SOLVE >**
09	Find All Anagrams in a String Pattern: Window frequency map	MEDIUM **PRACTICE**	**SOLVE >**
10	Subarray Sums Divisible by K Pattern: Prefix remainder map	MEDIUM **PRACTICE**	**SOLVE >**

Revision action: Select any 3 questions from this topic and solve them again without viewing your earlier solution.

08. Matrix

Interview focus: 2D traversal, transformations, markers and grid backtracking.
10 curated Easy-Medium practice questions

01	Set Matrix Zeroes Pattern: In-place markers	MEDIUM **PRACTICE**	**SOLVE >**
02	Spiral Matrix Pattern: Boundary traversal	MEDIUM **PRACTICE**	**SOLVE >**
03	Rotate Image Pattern: Transpose + reverse	MEDIUM **PRACTICE**	**SOLVE >**
04	Word Search Pattern: Grid backtracking	MEDIUM **PRACTICE**	**SOLVE >**
05	Flood Fill Pattern: Grid traversal	EASY **PRACTICE**	**SOLVE >**
06	Matrix Diagonal Sum Pattern: Diagonal indices	EASY **PRACTICE**	**SOLVE >**
07	Reshape the Matrix Pattern: Index conversion	EASY **PRACTICE**	**SOLVE >**
08	Toeplitz Matrix Pattern: Neighbour comparison	EASY **PRACTICE**	**SOLVE >**
09	Search a 2D Matrix II Pattern: Staircase search	MEDIUM **PRACTICE**	**SOLVE >**
10	Game of Life Pattern: In-place state encoding	MEDIUM **PRACTICE**	**SOLVE >**

Revision action: Select any 3 questions from this topic and solve them again without viewing your earlier solution.

09. Binary Search Tree

Interview focus: BST ordering, pruning, iterative traversal and updates.
10 curated Easy-Medium practice questions

01	Validate Binary Search Tree Pattern: Range constraints	MEDIUM **PRACTICE**	**SOLVE >**
02	Kth Smallest Element in a BST Pattern: Inorder traversal	MEDIUM **PRACTICE**	**SOLVE >**
03	Lowest Common Ancestor of a BST Pattern: Ordered navigation	MEDIUM **PRACTICE**	**SOLVE >**
04	Search in a Binary Search Tree Pattern: BST lookup	EASY **PRACTICE**	**SOLVE >**
05	Range Sum of BST Pattern: Pruned DFS	EASY **PRACTICE**	**SOLVE >**
06	Convert Sorted Array to Binary Search Tree Pattern: Balanced recursion	EASY **PRACTICE**	**SOLVE >**
07	Insert into a Binary Search Tree Pattern: BST insertion	MEDIUM **PRACTICE**	**SOLVE >**
08	Delete Node in a BST Pattern: BST deletion cases	MEDIUM **PRACTICE**	**SOLVE >**
09	Trim a Binary Search Tree Pattern: Range pruning	MEDIUM **PRACTICE**	**SOLVE >**
10	Binary Search Tree Iterator Pattern: Controlled inorder stack	MEDIUM **PRACTICE**	**SOLVE >**

Revision action: Select any 3 questions from this topic and solve them again without viewing your earlier solution.

10. Basic Dynamic Programming

Interview focus: State definition, recurrence, memoisation and tabulation.
10 curated Easy-Medium practice questions

01	Climbing Stairs Pattern: 1D DP	EASY **PRACTICE**	**SOLVE >**
02	Coin Change Pattern: Unbounded choice DP	MEDIUM **PRACTICE**	**SOLVE >**
03	Longest Increasing Subsequence Pattern: Sequence DP	MEDIUM **PRACTICE**	**SOLVE >**
04	Longest Common Subsequence Pattern: 2D sequence DP	MEDIUM **PRACTICE**	**SOLVE >**
05	Word Break Pattern: Prefix DP	MEDIUM **PRACTICE**	**SOLVE >**
06	House Robber Pattern: Take or skip	MEDIUM **PRACTICE**	**SOLVE >**
07	House Robber II Pattern: Circular take or skip	MEDIUM **PRACTICE**	**SOLVE >**
08	Decode Ways Pattern: Prefix decode DP	MEDIUM **PRACTICE**	**SOLVE >**
09	Unique Paths Pattern: Grid DP	MEDIUM **PRACTICE**	**SOLVE >**
10	Min Cost Climbing Stairs Pattern: Rolling state DP	EASY **PRACTICE**	**SOLVE >**

Revision action: Select any 3 questions from this topic and solve them again without viewing your earlier solution.

100-Question Progress Tracker

Topic	Starter Set	All 10	Timed Re-solve
Array	[ ] 7/7	[ ] 10/10	[ ]
String	[ ] 7/7	[ ] 10/10	[ ]
Stack	[ ] 1/1	[ ] 10/10	[ ]
Queue	[ ] 0/0	[ ] 10/10	[ ]
Searching	[ ] 2/2	[ ] 10/10	[ ]
Sorting	[ ] 3/3	[ ] 10/10	[ ]
Hashing	[ ] 3/3	[ ] 10/10	[ ]
Matrix	[ ] 4/4	[ ] 10/10	[ ]
Binary Search Tree	[ ] 3/3	[ ] 10/10	[ ]
Basic Dynamic Programming	[ ] 9/9	[ ] 10/10	[ ]

Interview reminder: The interviewer evaluates more than the final answer. Communicate the approach, time and space complexity, test cases, and code readability.

Keep Learning. Keep Building.

This practice guide is designed for Data Engineers preparing for early technical rounds where SQL and Easy-to-Medium problem-solving are evaluated together.

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Practice Platform

All question buttons in this guide open the corresponding LeetCode problem page for direct practice access.

LeetCode Problemset >

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